A MONSTER GAME THAT HAPPENS TO TEACH

# GEMMA MONSTERS

Five monsters hold a citadel. Each one feeds on a different way of being wrong.

> Every wrong answer in the game carries a **snare**:
> the exact wrong idea that makes that answer feel right.
>
> The monsters live on snares.
> You win by proving yours no longer works on you.

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|---|---|
| **What it is** | An offline tutor for Ontario's Grade 9 EQAO mathematics assessment, built as a game |
| **Where it runs** | One laptop. Gemma served locally through Ollama |
| **Event** | Build with AI — Gemma Hackathon, GDG Windsor |
| **Track** | Edge / On-Device |
| **Team** | Gokulakrishnan, Padmanabha, Taha, Naimah, Amarah |

**FRACTIS · EQUAZOR · STATIQ · POLYGOR · LEDGERLING**

and, above all five, the Collector

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No account. No internet during play. Nothing a student writes ever leaves the machine.

## 1 Enter the citadel

Every Grade 9 student in Ontario sits EQAO, the province's mathematics assessment, written once and marked province-wide. Around Windsor–Essex it is a date the whole house knows about. What is on offer is a tutor at sixty dollars an hour, or an app that puts a red X next to your answer and moves on.

The red X is the problem. A wrong answer in maths is almost never random. Ask a fourteen-year-old for $2/3 + 1/4$ and watch for $3/7$. Tops with tops, bottoms with bottoms. Tidy, symmetric, and wrong. Marking it wrong teaches nothing, because the student who wrote $3/7$ already believed $3/7$. That is not carelessness. It is a rule, applied faithfully. It is just the wrong rule.

We call each one a **snare**. A snare has a name, and once you can name it you can hunt it.

> **The rule of the game.** A monster sets a snare. You get caught. You beat the snare by answering fresh questions right *and* showing the path you took. Then the monster has nothing left to eat.

Five monsters guard the citadel, one for each strand EQAO assesses.

| MONSTER | GUARDS | LIVES ON |
|---|---|---|
| **Fractis** | Number | Fractions added straight across |
| **Equazor** | Algebra | A sign that slips crossing the equals |
| **Statiq** | Data | Mean and median blurred together |
| **Polygor** | Geometry & Measurement | A formula that is almost right |
| **Ledgerling** | Financial Literacy | Money and time, quietly mixed up |

Pick a monster and you drop into a torch-lit hall for five questions from the game's vault. Every wrong option on every one of those questions already has a snare written beside it. When you pick one, the game does not guess what went wrong in your head. It reads the label.

Pick a monster → Five questions from the vault → The snare that caught you most → Training, until it stops working → Back to the hall

## 2 One battle with Ledgerling

Ledgerling guards Financial Literacy and charges interest on every mistake. Here is one of its five questions, exactly as it sits in the vault.

> **FIN-Q1** Financial Literacy · budgeting and saving
>
> Priya has a part-time job in Barrie and earns \$800 each month. She saves 3% of her earnings every month. How much money will she have saved in total after 6 months?
>
> A. \$14,400.00 B. \$4,656.00 C. \$144.00 D. \$24.00

The answer is C, and the vault carries the working with it:

$$0.03 \times 800 = \$24 \text{ each month}, \qquad 6 \times 24 = \$144.$$

Say you pick D. You are not lost — you found \$24 correctly and then stopped. That option is labelled **Time-period mishandling**, and Ledgerling eats well tonight.

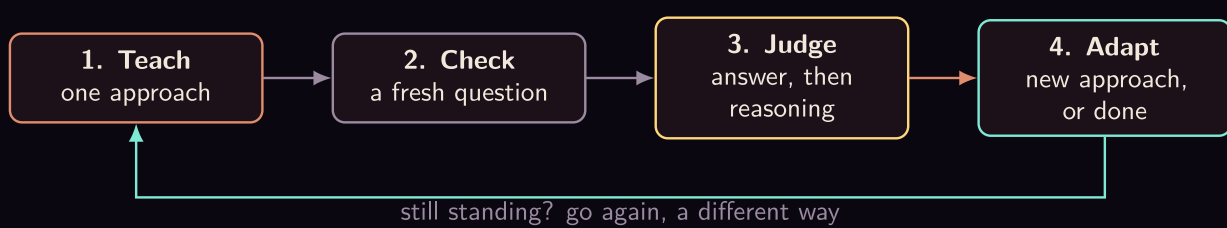
The other two wrong answers are different snares on the same question. A is **Percent–decimal conversion error**: multiply by 3 instead of 0.03 and you get $2,400 a month. B is **Wrong quantity reported**: $4,656 is the money Priya did *not* save. Three players can miss this one question for three different reasons and get three different evenings.

Across the five questions of the battle, the game keeps a tally. The snare that fed the monster most is the one it goes after first.


Nothing in that diagram is a judgement call. The labels come from the vault, the counting is arithmetic, and the biggest number wins. Gemma has not been asked anything yet.

## 3 How you beat a snare

One lucky answer proves nothing, so one lucky answer is not the bar. The loop runs like this, and it keeps running until you clear the bar or a cap stops it.




### 3.1 Four ways to explain, and no fifth

If one explanation does not land, the game does not repeat it louder. It changes the approach. There are exactly four, fixed in the code, and they are tried in this order unless Gemma argues for a different one.

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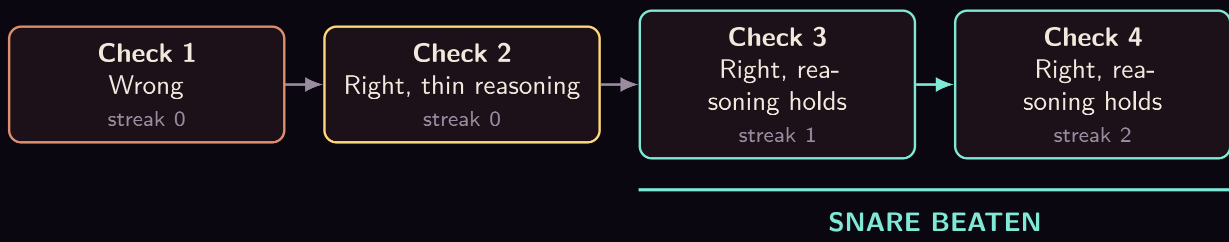
| **Direct correction** | Name the snare plainly — "you might think… but actually…" — state the right rule in two or three sentences, then work one example all the way through. |
| **Visual walkthrough** | A concrete model walked through one step at a time: a number line, an area model, groups of objects, a table. The point is to make you see why it works, instead of handing you another rule to remember. |
| **Side-by-side contrast** | The wrong method and the right method on the same problem, line against line, with a finger on the exact step where they split. |
| **Real-world analogy** | Money, pizza slices, game scores. Build the everyday version first, then map it back onto the maths. |

When you get a check question wrong, Gemma reads what you typed and picks which of the *remaining* approaches to try next, and says why in one sentence you can see. It cannot invent a fifth approach, cannot repeat one it has already used, and cannot leave the ladder. If its answer does not name a real approach, the code just takes the next rung and says so.

### 3.2 The bar, and the three ways a night ends




The bar is **two fresh questions right in a row**, with reasoning that holds up. The reasoning box under each answer is optional and the submit button never waits on it — but when you do write something, it counts:

- Right answer, reasoning that shows you understand: the streak goes up.
- Right answer, thin or circular reasoning: the streak stays where it is. You are not punished, but you have not shown anything yet.
- Right answer, reasoning that still contains the snare: the streak goes back to zero. You got there for the old wrong reason.
- Wrong answer: streak to zero, and a different approach next.

Three caps end the night, and all three are plain code: **four** check questions, **twelve** model calls, or a ladder with no rungs left. When one of them trips, the game stops and says which one. It does not drill you into the ground, and it does not pretend.

> **A fresh question is fresh, but it is not random.** Every follow-up is modelled on the exact question you just got wrong, and stays on that same topic. Not merely the same part of the course — slope from two points and the distributive property are both Algebra, and a player who could not solve the equation is not helped by coordinate geometry. If the vault has nothing left on that exact topic, the game would rather write a new question than hand you the wrong one.

## 4 What the vault holds

- **55 questions** across the five parts of the course: Number, Algebra, Financial Literacy, Geometry & Measurement, Data. The bank was audited against the published Grade 9 expectations and against what the EQAO assessment actually covers.
- **A worked solution on every single one.** That is what lets Gemma teach without being allowed to calculate.
- **A snare on every wrong option** — 165 of them — naming the faulty thinking that produces exactly that answer.
- **A balanced key**, 14/14/14/13 across A, B, C and D.
- **A topic on every question**, and the topic is a hard filter when the game hunts for your next one, not a preference.

Finding a follow-up is a tag lookup, not a search: the vault knows that NUM-6 and NUM-6b are the same idea at two depths, and it will take a sibling before it takes a stranger. If nothing on your topic is left, it returns nothing — and returning nothing is a real answer. The game then writes a question under audit, or stops and says the vault is out. Either beats practising the wrong thing.

## 5 Ten jobs for Gemma, one door

Gemma is not decoration on top of a quiz app. Take it out and there is no lesson, no judgement, no direction, and no letter home — just a marked test. Here is every job it does.

| | THE JOB | WHAT CODE STILL OWNS |
|---|---|---|
| 1 | **Writes the lesson** | The approach, and the verified working it may not contradict |
| 2 | **Explains why your wrong answer felt right** | The right answer; it may not redo the arithmetic |
| 3 | **Gives a hint without giving it away** | How deep the hint may go |
| 4 | **Judges whether your reasoning holds up** | Three labels and no fourth; whether the answer was right |
| 5 | **Chooses the next approach, and says why** | The four approaches, and the order if the reply is unusable |
| 6 | **Writes a fresh question when the vault is out** | The topic, and a refusal if the question drifts off it |
| 7 | **Solves that question again, blind** | Whether the two answers match, and what happens if not |
| 8 | **Decides which fight comes next** | The list it may choose from: real fights, still open |
| 9 | **Speaks as the citadel — taunts, reactions, relics** | That none of it touches your score |
| 10 | **Writes the note that goes home** | Every single figure on that page |

Read that table as one evening rather than a list. You miss a question, so it explains why your answer felt right (2). It teaches the idea properly (1). The vault has nothing left on your topic, so it writes you a fresh one (6) and then solves that question again, blind, to check its own key (7). You answer and you type why — and it decides whether that reasoning actually holds (4). It does not, quite, so it picks a different way to explain and tells you which and why (5). Then the night ends and it writes home (10).

Two of those jobs turn up twice over. Number one also works down at the Collector's training ground, where Gemma whispers the mental shortcut before you go in; number four works there too, reading the exact questions you missed afterwards and naming the one pattern in them. Same door, same caps, different room.

**One door, and what it buys.** Every prompt and every reply in this game passes through a single function. That is why the caps can be counted at all. It is why the whole app moves to a bigger or a smaller Gemma by changing one environment variable, with nothing else to edit. And it is why a model that is slow, busy, or simply not installed makes the game go quiet rather than crash — the door hands back clearly marked placeholder text, and the battle, the marking and the vault carry on without it.

## 6 True and judged

Here is the line the whole design is built on. Code owns what is **true**. Gemma owns what is **judged**. Nothing crosses.

**CODE OWNS WHAT IS TRUE**

Every answer key
The snare on each wrong option
Every number that reaches a page
The lists it may choose from
The streak, the caps, the stop

**Gemma OWNS WHAT IS JUDGED**

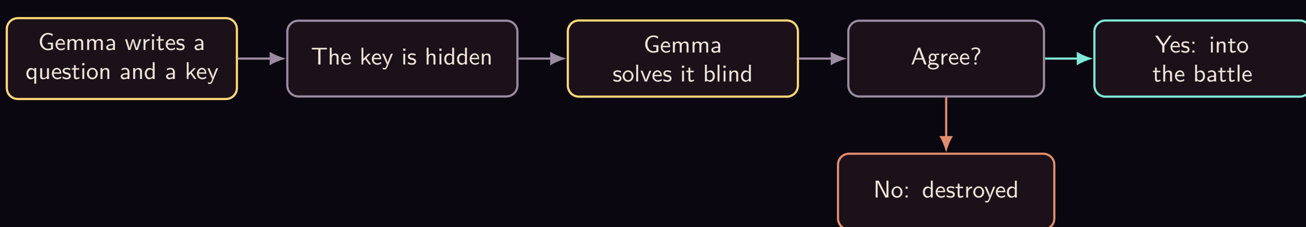
How to explain this snare tonight
Whether your reasoning holds up
Which approach is worth trying next
Which fight this run has earned
What all of it means, for a parent

Every one of those left-hand rows is there because we watched the right-hand side get something wrong first. Four shields came out of that.

**Shield 1 — Gemma never redoes the maths.**
Asked to re-explain a question it had already solved, it would quietly redo the sums and land somewhere new, in confident prose, right next to the correct answer. So the verified working now rides inside the prompt as the only arithmetic it is allowed to state. Its job is to explain *why* your method fails, never to recompute.

**Shield 2 — the snare is read, not guessed.**
Every wrong option in the vault carries the faulty thinking that produces it. When you pick one, the game looks the snare up. No model call, no inference, no chance of naming the wrong trap and teaching you a lesson you did not need.

**Shield 3 — a written question must survive itself.**
We caught the model writing a perfectly good question and marking the wrong option as the answer. A student would have been told they were wrong for being right. So now anything Gemma writes has to solve itself again *blind* — without seeing the key it just wrote — and the two answers have to match. They disagree, the question is destroyed and nobody ever sees it.



three tries, then the loop stops and says so

**Shield 4 — Gemma may not put a number on the page.**
It once wrote "beaten three tricks" when one had been beaten, borrowing a 3 that belonged to a different count on the same screen. Checking figures one at a time cannot catch a real number attached to the wrong noun. So the rule is absolute: on the page that goes to a parent, code prints every figure and Gemma prints none. Any reply containing a number — in digits or spelled out — is thrown away and a sentence built from the same facts goes out instead.

There is a fifth guard that is not about the model at all. In an early draft of the vault, 58% of the correct answers sat on option A. A player who always guessed A would have scored well and learnt nothing.

The key is now balanced across all four letters, and every note about a wrong answer is keyed to the answer's *text*, so it stays true however the options are shuffled.

Underneath all of it, one sentence: **a small model is a capable judge and an unreliable authority.** Everything Gemma says in this game is a judgement. Nothing it says is the source of what is mathematically true.

## 7 When the snare holds

Some nights you do not beat it. The game does not keep grinding. It says the battle is ending, why it is ending, and that a person is probably more use than another question right now. A note goes home naming the approaches already tried, so nobody starts from scratch.

That is also when the Collector notices you. He is the giant skull above the citadel, and the five monsters are afraid of him. He does not test new ideas. He tests the arithmetic you are supposed to already own, and he tests it against a clock: **ten questions, eight seconds each, three lives**. Nothing in his arena counts against your quiz record.

Lose to him and you go down to his lieutenants, one mental shortcut each.

| | |
|---|---|
| **Twinfang** | Doubling and halving. Double by adding a number to itself; multiply by 4 by doubling twice. |
| **The Niner** | Nines, fast. Multiply by 10, then subtract the number once. |
| **Splitjaw** | Making tens. $47 + 38$ becomes $47 + 3$, then $+35$. |

Each drill is ninety seconds on three lives. Gemma whispers the shortcut before you go in, reads the exact questions you missed afterwards, and picks the one trick worth drilling next. Then it chooses which lieutenant is worth your time — from the list of the ones you have actually fought, with the scores you actually got, and it names that evidence out loud when it sends you.

Retreating from any fight is free. The monsters remember it, and they will mention it when you come back.

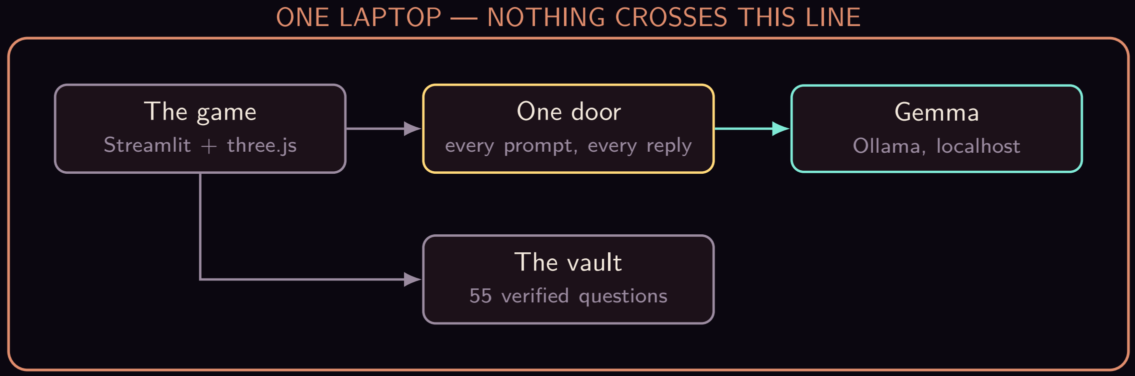
## 8 A letter reaches home

After a battle, a note is written for whoever is at home. Not only on the bad nights — a parent who only hears from us on the worst day cannot see a pattern, so notes go after any battle with mistakes in it, after a hand-off, and after a win.

The note says which monster was fought, which snare caught you most, which explanations were tried, whether the snare fell or is still standing, and three things to try at the kitchen table with coins, cups, food or paper. Ten minutes, no jargon, nothing to buy. And it is about one evening — it is not a label anyone gets to keep.

One button on that page prints a practice sheet: up to ten questions on that one snare, room to write under each, and the answer key on its own page so a parent can hand over the questions without handing over the answers. Vault questions first, because those come with working a parent can check. Anything Gemma writes to fill the sheet has already solved itself blind, the same as on screen. If it cannot fill ten, the sheet comes out short. It never comes out with a question we could not verify.

## 9 It really runs, and it runs on one laptop




The whole game exists and you can play it end to end today: the opening scene, the citadel you can orbit, five monster halls, the training loop, the Collector and his three lieutenants, the notes home and the printable sheets, and the gate that opens when the last seal breaks.

No account. No sign-in. No internet needed once it is installed. Three.js is vendored into the repository, the monster models are CC0, the audio is ours, and there is not a single request to a CDN. A fourteen-year-old's worked-out mistakes are exactly the kind of thing that should not be uploaded anywhere, so they are not. They stay in a file on the family's own machine.

And if Gemma is not installed at all, the app still opens, still runs the quiz, still marks it, and says plainly where the writing would have been.

## 10 What we have not shown

No student has used this yet. Not one.

Nothing in the game measures whether any of it sticks. Beating a snare means two fresh questions right in a row with reasoning that held up — that is our bar, chosen by us, and it is a claim about one evening, not about next month. We have not tested whether a snare beaten on Tuesday is still beaten in June. We have not watched a real fourteen-year-old read a lesson and decide whether it made sense. We do not know which monster they will pick first, or which explanation actually lands.

Those are the next things to find out, and they need students in a room, not more code. Until then, what we have is this: a game that runs offline on one laptop, that names the specific wrong idea behind a wrong answer instead of marking it and moving on, and that is honest about which parts of it are true and which parts are judgement.